GUNRAAJ SINGH

Software Engineering | Embedded Systems | HIL Automation | EV Modelling

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Education

B.Tech in Electrical Engineering — **Expected May 2027**
Pandit Deendayal Energy University (PDEU) — *Gandhinagar, India*
CGPA: 8.26 / 10 | **Relevant Coursework:** Machine Learning, Control Systems, Power Electronics, Electric Drives, Electric Vehicle Technology, Engineering Mathematics

Technical Skills

Languages: Python, C++, C, SQL, MATLAB
Software & CS: OOP, Data Structures & Algorithms, multithreading, PySide6 / Qt, NumPy, Pandas, OpenCV, Git
Embedded & Automotive: ESP32, FreeRTOS, PlatformIO, CAN / TWAI, DBC, cantools, UART, GPIO, PWM
Modelling & Testing: MATLAB / Simulink, Simscape, HIL automation, serial protocols, signal validation, regression testing

Experience

BGauss Auto Private Limited — **Jun 2026 – Jul 2026**
R&D Intern, Software & Embedded Systems — *Pune, India*

- Built an **end-to-end HIL validation platform** linking a **Python / PySide6** desktop application with **C++ / FreeRTOS** ESP32 firmware for repeatable EV electronics testing over USB serial and CAN / TWAI.
- Designed a no-code **visual test sequencer** with non-blocking execution, reusable actions, relay and PWM control, hardware profiles, pin-safety checks, structured session logs, and automated reports.
- Integrated **JSON-over-Serial** and DBC-based CAN signal decoding with `cantools`; implemented assertions, wait / timeout conditions, and sampled **MIN / MAX / MEAN** validation.
- Audited and extended a **MATLAB / Simulink EV powertrain model** covering road-load physics, battery SOC, regenerative braking, Wh/km, range estimation, and motor sizing; benchmarked the Indian Drive Cycle against MoRTH data.

Selected Projects

Vibration Analysis & Diagnostic System — *Python, OpenCV, Signal Processing*
Built a vision-based analyser using Lucas–Kanade optical flow to track **100+ structural features**; applied FFT and Butterworth filtering to estimate dominant vibration frequencies from video.

Synchronous Buck Converter (24 V → 12 V) — *STM32, Embedded C, Power Electronics*
Designed a **40 kHz synchronous buck converter** using an STM32G030, IR2113 gate driver, and IRFZ44N MOSFETs; implemented **cascaded dual-loop PI control** with ACS712 current sensing and a hand-wound 300 µH inductor.

ANN-Based PM-BLDC Speed Control — *MATLAB / Simulink, Neural Networks*
Designed and evaluated a **3-8-1 ANN controller** trained on approximately **25,000 samples** using Levenberg-Marquardt optimisation; analysed transient response, duty-cycle saturation, and steady-state behaviour.

Leadership

Anirveda – The Techno-Economics Club, PDEU — **Jun 2025 – Jun 2026**
Vice President — *Gandhinagar, India*

- Led a **30+ member** student organisation and coordinated **10+ events** reaching **500+ participants**; drove sponsorship outreach and mentored **15+ junior members**.